Year 11 Mathematics

Handling Data 4

4.5 Averages for Discrete Data

Averages for Discrete Data

STARTER · 5 MIN

Date: _____ **Title:** Averages for Discrete Data

LEARNING OBJECTIVES

- Find the mean, median, mode and range of a list
- Find averages from a frequency table
- Choose the best average and solve problems

STARTER *Answer in your book. No calculators.*

1. Order 7, 3, 9, 1, 5
2. Work out 4 + 8 + 6 + 2
3. Work out 20 ÷ 4
4. Work out 3 × 4 + 2 × 5

Starter — Answers

STARTER

Tick or correct in green pen

1. **1, 3, 5, 7, 9**

2. **20**

3. **5**

4. $12 + 10 = \mathbf{22}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the mean, median, mode and range of a list	☐
SILVER	I can find averages from a frequency table	☐
GOLD	I can choose the best average and solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Averages and Range

INTRODUCTION · 3 MIN

An average is a typical value. The range shows how spread out the data is.

Mean

Total ÷ number of values.

(2 + 4 + 6) ÷ 3 = 4

Median

Middle value when in order.

1, 3, 5 → 3

Mode

Most common value.

2, 3, 3, 5 → 3

Range

Largest − smallest.

9 − 1 = 8

Examples 1 and 2 — Averages of a List

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint mean = $\frac{\text{total}}{\text{number of values}}$

Example 1

Find the mean and range of
4, 7, 3, 8, 3

Example 2

Find the median and mode of
6, 2, 9, 2, 5, 8

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Total = 25

Mean = 25 ÷ 5

= 5

Range = 8 − 3

= 5

Tip: range is one number, not 3–8.

Example 2

2, 2, 5, 6, 8, 9

Median = (5 + 6) ÷ 2

= 5.5

Mode **= 2**

Tip: put the numbers in order first.

Task 1

PRACTICE · 9 MIN

BRONZE *Answer in your book*

Hint $\text{mean} = \dfrac{\text{total}}{\text{number of values}}$

1. Mode of 4, 6, 6, 2, 7
2. Range of 12, 5, 9, 15, 7
3. Median of 8, 3, 6, 1, 9
4. Mean of 5, 8, 2, 9
5. Median of 4, 10, 7, 2, 8, 6

★ Mean of 2.5, 3, 4.5, 6

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. **6**

2. $15 - 5 = \mathbf{10}$

3. 1, 3, 6, 8, 9, Median = **6**

4. $24 \div 4 = \mathbf{6}$

5. 2, 4, 6, 7, 8, 10, $(6 + 7) \div 2 = \mathbf{6.5}$

★ $16 \div 4 = \mathbf{4}$

Examples 3 and 4 — Averages from a Frequency Table

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint mean = $\frac{\Sigma fx}{\Sigma f}$

Example 3

Find the mean score.

Score	Frequency
1	3
2	5
3	2

Example 4

Find the median and mode.

Score	Frequency
1	3
2	5
3	2

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$1 \times 3 + 2 \times 5 + 3 \times 2 = 19$

Total frequency $= 10$

Mean $= 19 \div 10$

$= \mathbf{1.9}$

Tip: divide by the total frequency, not 3.

Example 4

10 values: 5th and 6th

Both are 2

Median = 2

Mode **= 2**

Tip: the mode has the highest frequency.

Task 2

PRACTICE · 9 MIN

SILVER

Use the frequency table of goals

Hint mean = $\frac{\Sigma fx}{\Sigma f}$

1. Write down the mode
2. How many matches were played?
3. Find the median
4. Find the total number of goals
5. Find the mean

Goals	Frequency
0	4
1	7
2	5
3	3
4	1

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. **1 goal**

2. $4 + 7 + 5 + 3 + 1 = \mathbf{20}$

3. 10th and 11th values, **1 goal**

4. $0 + 7 + 10 + 9 + 4 = \mathbf{30}$

5. $30 \div 20 = \mathbf{1.5}$

Examples 5 and 6 — Problems and Choosing an Average

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint total = mean × number of values

Example 5

(3 marks)

The mean of 5 numbers is 8. Four of them are 6, 9, 7, 10. Find the fifth number.

Example 6

(2 marks)

Salaries: £20k, £22k, £21k, £23k, £95k. Which average is best? Explain.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$\text{Total} = 5 \times 8$$
$$= 40$$
$$6 + 9 + 7 + 10 = 32$$
$$\text{Fifth} = 40 - 32$$
$$\mathbf{= 8}$$

Tip: mean × number = total.

Example 6

£95k is an outlier

It makes the mean too big

Use the median: £22k

Tip: outliers affect the mean.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint total = mean × number of values

1. The mean of 4 numbers is 12. Three are 10, 15 and 9. Find the fourth. (3 marks)

2. The mean of 6 numbers is 5. The number 12 is added. Find the new mean. (3 marks)

3. Shoe sizes: 5, 6, 6, 7, 6, 8, 12. Which average is best? Why? (2 marks)

4. Class A: mean 64, range 30. Class B: mean 70, range 12. Compare the classes. (2 marks)

5. Values 1, 2, 3, 4 have frequencies 2, k, 4, 1. The mean is 2.5. Find k. (4 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $4 \times 12 = 48, \quad 48 - 34 = \mathbf{14}$

2. $6 \times 5 = 30, \quad 42 \div 7 = \mathbf{6}$

3. 12 is an outlier, Median $= \mathbf{6}$

4. B higher on average, **B more consistent**

5. $18 + 2k = 2.5(7 + k), \quad 0.5 = 0.5k, \quad k = \mathbf{1}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the mean of 3, 7, 8

SILVER Find the median of 9, 2, 5, 4

GOLD The mean of 3 numbers is 10. Two are 8 and 15. Find the third.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $18 \div 3 = \mathbf{6}$

SILVER 2, 4, 5, 9, $(4 + 5) \div 2 = \mathbf{4.5}$

GOLD $30 - 23 = \mathbf{7}$

Exam Practice — Averages of a List

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint $\text{mean} = \dfrac{\text{total}}{\text{number of values}}$

Question 1

(3 marks)

Here are 7 numbers.

5 8 3 8 6 10 2

(a) Write down the mode. (1)

.....

(b) Work out the median. (1)

.....

(c) Work out the range. (1)

.....

Question 2

(2 marks)

Work out the mean of

12 15 9 14 10

.....

Exam Practice — Averages from a Frequency Table

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint mean = $\frac{\Sigma fx}{\Sigma f}$

Question 3

(4 marks)

Goals scored in 20 matches.

Goals	0	1	2	3	4
Frequency	4	6	5	3	2

(a) Write down the mode. (1)

.....

(b) Work out the median. (1)

.....

(c) Work out the mean. (2)

.....

Question 4

(3 marks)

The table shows the number of pets of 20 students.

Work out the mean.

Pets	0	1	2	3
Frequency	7	9	3	1

.....

Exam Practice — Problems with Averages

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint total = mean × number of values

Question 5

(3 marks)

The mean of 6 numbers is 9.

One number is removed.

The mean of the other 5 numbers is 8.

Which number was removed?

.....

Question 6

(4 marks)

Values 1, 2, 3, 4 have frequencies 3, x, 5, 2.

The mean is 2.5

Work out x.

x = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 8 — B1

(b) 2, 3, 5, 6, 8, 8, 10: 6 — B1

(c) 10 − 2 = 8 — B1

Tip: put the numbers in order first.

Question 2

12 + 15 + 9 + 14 + 10 = 60 — M1

60 ÷ 5 = 12 — A1

Tip: divide the total by how many numbers.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) 1 — B1

(b) 10th, 11th: 1, 2 → 1.5 — B1

(c) 0 + 6 + 10 + 9 + 8 = 33 — M1

33 ÷ 20 = 1.65 — A1

Tip: the mean is Σfx ÷ Σf.

Question 4

0 + 9 + 6 + 3 = 18 — M1 M1

18 ÷ 20 = 0.9 — A1

Tip: divide by 20 students, not 4 rows.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$6 \times 9 = 54$	M1
$5 \times 8 = 40$	M1
$54 - 40 = 14$	A1

Tip: total = mean × number of values.

Question 6

$\Sigma f = 10 + x,\ \Sigma fx = 26 + 2x$	M1
$26 + 2x = 2.5(10 + x)$	M1
$26 + 2x = 25 + 2.5x$	M1
$x = 2$	A1

Tip: form an equation from Σfx = mean × Σf.